

LESSON GUIDE

Technology Plus

Unit 4: Networking

2.9.1 – Small Wireless Network Basics

Lesson Overview:

In this lesson, students will explore the core functions and performance factors of small wireless networks, including how frequency bands, interference, and emerging Wi-Fi standards impact real-world connectivity.

Students will:

- Describe how wireless networks transmit data and what factors affect their performance.
- Compare the characteristics of 2.4 GHz, 5 GHz, and 6 GHz frequency bands.
- Explain how improvements in wireless standards enhance speed, efficiency, and device support in modern networks.

Guiding Question: Why does Wi-Fi work great sometimes and terribly other times – even on the same device?

Suggested Grade Levels: 8-12

Technology Needed: Yes

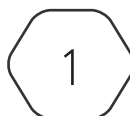
Approximate Time Required: 75-90 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.9 – Explain the basic capabilities of a small wireless network.
 - 802.11n/ac/ax
 - Speed considerations
 - Interference and attenuation factors
 - Older vs. newer standards
 - Band options
 - 2.4 GHz
 - 5 GHz
 - 6 GHz

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Lesson Background

Background Information:

Wi-Fi isn't magic – it's math, physics, and strategy packed into a network most students use daily without ever thinking about it. This lesson breaks open the invisible systems behind that everyday experience, helping students recognize how distance, interference, data size, and device type all impact performance. By demystifying why some connections are slow, glitchy, or perfect, students gain real-world insight into how small wireless networks work – and how to troubleshoot or optimize them. It's the first step in helping them move from users to informed participants in modern digital infrastructure.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 2.9.1 – Small Wireless Network Basics • Activity Slides 2.9.1 – Wi-Fi Wars: Battle of the Bands • Activity 2.9.1 – Wi-Fi Wars 2.4 GHz Device Cards • Activity 2.9.1 – Wi-Fi Wars 5 GHz Device Cards • Activity 2.9.1 – Wi-Fi Wars 6 GHz Device Cards • Activity 2.9.1 – Wi-Fi Wars Walls • Activity 2.9.1 – Wi-Fi Wars Event Calls strips • String, cones, or tape to mark band lanes • Bag/bowl/envelope to hold strips • Countdown timer
Materials per Student	• Student Notes 2.9.1 – Small Wireless Network Basics • Student Handout 2.9.1 – Small Wireless Network Basics • Assessment 2.9.1 – Small Wireless Network Basics (optional)

Lesson Background

Cyber Terms:

Term	Definition
wireless network	a network that connects devices using radio waves instead of cables
speed	how much data can move across the network in a certain amount of time, usually measured in megabits per second (Mbps) or gigabits per second (Gbps)
latency	the delay between sending a request and getting a response, measured in milliseconds (ms)
attenuation	the weakening of a wireless signal as it travels or passes through walls, floors, or other barriers
interference	disruption of Wi-Fi signal by other signals, like microwaves, Bluetooth devices, or nearby networks
band	a specific range of radio frequencies (like 2.4 GHz or 5 GHz) used for wireless communication
bandwidth	the amount of data that can pass through a network at once; more bandwidth means more potential speed
multiple input, multiple output (MIMO)	a router that uses multiple antennas to send and receive more data at the same time
multi-user MIMO	a device that lets wireless devices connect to a network, usually by adding Wi-Fi to a wired system
(MU-MIMO)	a router that sends data to multiple devices at once instead of taking turns
orthogonal frequency division multiple access (OFDMA)	Wi-Fi channels are split into smaller parts to handle more devices at the same time
target wake time (TWT)	devices schedule when to wake up and check in with the network, which saves battery life

2.9.1 – Small Wireless Network Basics

Guiding Question: Why does Wi-Fi work great sometimes and terribly other times – even on the same device?

Lesson Launch (5 minutes)

- At Slide 3, **say:** “What if I told you this was happening all around you right now? You can’t see it, but the air above your head is packed with tiny packets of data – your homework, your Spotify playlist, your grandma’s FaceTime call – all trying to move through the same airspace.”
- **Ask:** “Look around the room. How many people are in here?”
- **Ask:** “Now guess: how many devices do you think are connected to this school’s Wi-Fi network right now? Not just ours – I mean the whole building.”
- **Say:** “Even if it’s just *two devices per person* – a Chromebook and a teacher laptop – you’re already looking at hundreds. Add in smart boards, printers, security cameras, admin computers, backup systems... and suddenly your Wi-Fi network is packed like rush hour on the I-5 in Los Angeles – during a Dodgers game – with a wreck – in the rain.”
- **Say:** “Most of those devices are crowding into just a couple of lanes – especially the 2.4 GHz band. Today, we’re going to learn how bands, bandwidth, and better tech help break the traffic jam.”

Bandwidth Breakdown (25-30 minutes)

What is a Wireless Network?

- Distribute Student Handout 2.9.1 – Small Wireless Network Basics and encourage them to fill in answers as you go through the lesson.
- Start here by asking, “How many of you have connected to Wi-Fi today?” You’ll get every hand. Now flip it: “What’s actually happening when you do that?”
- Emphasize **mobility and flexibility** – this is why students can work on a laptop in bed, stream on a bus, or text from the bathroom.
- The asterisk is performance: **Mobility comes at a price.** The rest of the lesson is about what that price is and how to manage it.

Speed vs. Latency

- Accentuate the difference: **Speed** is about how *much* data can go; **latency** is about how *soon* it gets going.
- Use the **highway vs. on-ramp** analogy:
- **Speed** = How many lanes on the highway
- **Latency** = How long the light stays red at the on-ramp
- **Ask:** “Which matters more for Zoom calls and gaming?”

- Students should say latency and recognize that we want to avoid lag with video calls and gaming.

Speed Considerations & Latency Ratings

- This is where the tech terms become tools. Use examples:
- **Channel width:** “Wider roads let more cars through. 80 MHz is a big highway. 20 MHz is a side street.”
- **Spatial streams:** “Think of multiple antennas as multiple mouths talking at once – or multiple ears listening.”
- Highlight that **newer Wi-Fi handles busy spaces better** – not because of more power, but smarter tech.

Attenuation and Interference

- Let students brainstorm what gets in the way of Wi-Fi at home (you’ll get microwaves, fish tanks, walls, siblings streaming TikTok).
- Reinforce that 2.4 GHz has reach, but it’s also the most congested frequency band, used by many household devices – even microwaves and smart coffee makers.
- Demo tip: Put your phone on airplane mode, then turn Wi-Fi back on and try to connect to a network – it shows just how many signals are overlapping around us.

Wi-Fi Standards Timeline

- This slide is not for memorizing – it’s for recognizing **evolution**.
- **Ask:** “Why do we keep inventing new Wi-Fi versions?”
Students may say, “To fix the problems the last version couldn’t solve.”
- Point out that each new version:
 - **Gets faster**
 - **Handles more devices**
 - **Adds smarter features**
- Optional fun fact: Wi-Fi 0 didn’t support encryption. That’s right – *open* by default.

Bands and Bandwidth

- Reinforce the metaphor:
 - **Band = where the road is** (neighborhood, highway, tollway)
 - **Bandwidth = how wide the road is**
- Great moment to point out that more **bandwidth isn’t always better** if your device or environment can’t handle it (e.g., wide roads are useless if the car can’t go over 25 miles per hour).
- Band Options
- Focus on the **tradeoffs:**
 - 2.4 GHz = slow but reliable
 - 5 GHz = good mix of speed and reliability
 - 6 GHz = blazing, but needs a clear path and a new car

Comparing Standards & Efficiency Features

- This slide highlights key efficiency features – direct students to focus on how each one improves wireless performance:
 - **MIMO:** More antennas = more ways to talk.
 - **MU-MIMO:** Talk to *multiple people at once*.
 - **OFDMA:** Efficient carpooling – no wasted space on the bus.
 - **TWT:** Devices only wake up when they need to.
- Point out how modern Wi-Fi is optimized for the “**smart home**” lifestyle – where dozens of little devices all need a little data but don’t need constant attention.

Activity: Wi-Fi Wars – Battle of the Bands (35-45 minutes)

Note: This simulation runs best when treated like a lab, not a competition. The goal isn’t to win – it’s to experience the conditions that affect Wi-Fi performance.

If you don’t have a full class of 36 students, repurpose any unused **2.4 GHz device cards** by taping them to chairs or placing them along the 2.4 GHz lane. These act as “phantom devices” to simulate **frequency congestion** from other connected devices (like smart TVs, Bluetooth gadgets, or smart plugs). They don’t move but still crowd the lane – just like in real networks! You could also just print another set of the 2.4 GHz cards to make sure you have enough devices for students and phantom devices!

Prep: Print the Activity 2.9.1 – Wi-Fi Wars Device Cards on colored paper (red = 2.4 GHz, blue = 5 GHz, green = 6 GHz) then cut out to give one to each student. Use the string, cones, or tape to mark band lanes. The Activity 2.9.1 – Wi-Fi Wars Walls handouts could be printed on the same color paper as device bands to help you keep straight which walls go in which lane, but plain paper is fine. Print the 2.9.1 – Wi-Fi Wars Event Calls strips, cut them out, and place them in a bag/bowl/envelope to have on hand during play.

Objective: Help students visualize and simulate how different Wi-Fi bands, bandwidth, interference, and distance impact performance.

Storyline: Each student is a data packet from their assigned device trying to reach a destination (like loading a website or sending a message).

Roles:

- **Data Packets (students):** carry messages across the room
- **Router (stations):** act as the starting point
- **Destination Server:** the endpoint that students are trying to deliver their packet to (themselves)

Obstacles: Place the walls at the same point in the path for all bands. Each band responds to obstacles differently – students follow the instructions printed on their lane’s wall cards.

Event Calls: To simulate real-world interference, routing issues, and network congestion, introduce Event Calls at any point during gameplay.

- Some events take effect **immediately mid-turn** (e.g., “Microwave on!” makes 2.4 GHz students freeze instantly)
- Others take effect **starting on the next turn** (e.g., “Bluetooth speaker nearby!” means no move next round)
- A few are **global** and impact all students (e.g., “Router reboot!”)
- **Pro Tip:** You can draw an Event Call before the game begins to kick things off or pull one every 1–2 turns depending on the chaos level you want.

How it works:

- **Set up 3 zones**, each one tied to a different band (2.4, 5, 6 GHz). Each band lane should have a designated endpoint labeled “**Destination Server**.”
- **Assign students as devices** with:
 - Band assignment (card color)
 - Data Size (Small/Medium/Large data = affects how fast they can go)
 - Device Type (old device = no access to 6 GHz)

Game Play: Turn-Based Movement System

- To simulate packet speed differences based on data size, the teacher leads the class in structured turns.
- Each turn lasts for a 6-Mississippi count. Use the countdown timer on your phone, computer, or smart board to count down six seconds.
- The teacher calls: “Begin Turn!” and then presses start on the timer.
- Students move according to their Data Size and Band:
 - 2.4 GHz – Slowest
 - Small data size: 1 step per Mississippi (6 steps total)
 - Medium data size: 1 step every 2 Mississippi (3 steps)
 - Large data size: 1 step every 3 Mississippi (2 steps)
 - 5 GHz – Faster
 - Small data size: 2 steps per Mississippi (12 steps total)
 - Medium data size: 1 step per Mississippi (6 steps)
 - Large data size: 1 step every 2 Mississippi (3 steps)

- 6 GHz – Fastest
 - Small data size: 3 steps per Mississippi (18 steps total)
 - Medium data size: 2 steps per Mississippi (12 steps)
 - Large data size: 1 step per Mississippi (6 steps)
- Students may also switch bands as their cards allow, but that must be made laterally (without advancing forward). They should switch bands, then continue their forward progression. **Once a student changes bands, they must follow the movement speed rules for their new band – even mid-turn.**
- When the timer sounds, the teacher calls: "Turn Over!" and all students freeze in place.

Tips to Manage Chaos:

- **Build anticipation:** Say, "Today, you're going to become the traffic on a Wi-Fi network – let's see how distance, interference, and data size actually affect your connection."
- **Use a visible countdown:** Show that turns are timed and the rounds are deliberate – not rushed. Even better if the timer is projected.
- **Assign a student helper** to draw event calls or help reset between rounds – spreads the load and keeps engagement high.
- **Post-play pause:** Take a moment to say, "Okay, let's breathe – that was a lot of movement, but here's why we did it..."

Wrap-Up Discussion

- Which band handled the most traffic?
Usually 2.4 GHz – it supported the most devices and had the longest range but also got crowded and slowed down by interference.
- Which band was the fastest?
6 GHz – it allowed the most steps per turn for small and medium packets. Large packets still moved quickly, unless they were blocked by a wall.
- Where did interference have the biggest impact?
2.4 GHz – it was hit most often by event calls like "Microwave on!" and had the most overlap with nearby devices.
- What's the tradeoff between range and speed?
2.4 GHz reaches far but is slower and more interference-prone. 6 GHz is blazing fast but can't get through walls and has the shortest reach. 5 GHz sits in the middle.
- How did band choice affect performance?
Students in higher bands (5 and 6 GHz) moved faster but had to make strategic choices about lane switching and obstacle handling. Students on 2.4 GHz could keep going, even through walls, but at a slower pace.

- How did data size affect how quickly packets reached the destination?
Small packets always finished first, especially in higher bands. Large packets took more time, especially if they were in slower bands or got stuck at walls.
- What band would you pick for:
 - A smart thermostat?
2.4 GHz – doesn't need speed, just consistent connection across the house.
 - A gaming console?
5 GHz or 6 GHz – needs speed and low latency and usually stays near the router.
 - A phone in a crowded coffee shop?
6 GHz if supported – it avoids congestion. Otherwise, 5 GHz is a solid choice for fewer dropouts.

Putting It All Together (10 minutes)

- **Say:** "Today we explored why Wi-Fi works great sometimes and terribly other times – even on the same device."
- **Ask:** "What surprised you most about how wireless networks actually work?"
I didn't realize how many things can interfere with Wi-Fi – like microwaves and Bluetooth. I just thought it was either "working" or "not working."
- **Ask:** "How would you explain the difference between speed and latency to someone at home?"
Speed is how much data can move at once, but latency is how fast it starts moving. It's like having a big highway (speed) but needing a green light to actually go (latency).
- **Ask:** "If you were designing a smart home, which Wi-Fi band would you use for each of these: a gaming console, a smart doorbell, and a tablet?"
Gaming console: 5 or 6 GHz for low lag;
Smart doorbell: 2.4 GHz for long range;
Tablet: 5 GHz for good speed and okay range.
- **Ask:** "What advice would you give someone who's always complaining their Wi-Fi is slow?"
I'd tell them to check if they're on the right band and to see if there are a bunch of other devices connected. Also maybe move closer to the router or switch to 5 GHz if they can.
- **Say:** "The next time someone says their Wi-Fi is slow, you'll know it's not just bad luck – it's bandwidth, bands, and a whole bunch of invisible traffic. Now you speak the language of networks."